

ORSO analysis working group update

2026-07-24

SXNS / ORSO Satellite Meeting

Current chairs

- Andrew Nelson (ANSTO, Australia)
- Brian Maranville (NIST, USA)

Continued curation of validation suite

- <https://github.com/reflectivity/analysis>
- Checks that all software packages calculate reflectivity correctly (smeared/unsmeared calculations)

 VALID: checking BornAgain ORSO Val. refnx #352: Pull request #86 synchronize by andyfaff	andyfaff:ba	 2 minutes ago  55s	...
 VALID: checking BornAgain ORSO Val. GenX #352: Pull request #86 synchronize by andyfaff	andyfaff:ba	 2 minutes ago  1m 15s	...
 VALID: checking BornAgain Code Formatting Check #73: Pull request #86 synchronize by andyfaff	andyfaff:ba	 2 minutes ago  25s	...
 VALID: checking BornAgain ORSO Val. BornAgain #354: Pull request #86 synchronize by andyfaff	andyfaff:ba	 2 minutes ago  38s	...
 VALID: checking BornAgain ORSO Val. Refl1D #138: Pull request #86 synchronize by andyfaff	andyfaff:ba	 4 minutes ago  45s	...

Validation data: provenance

- Added top-level documentation in `validation` folder to describe generation of resolution-smear data
- Identified sources of mismatch between refl1d smearing algorithm and refnx (different effective integration cutoff)

Code Signing

- User-installable packages: OS prevents unsigned code installs
- Many reflectometry packages signed using International Scattering Alliance certificate (for distribution)
 - GenX, Refl1D, refnx ...
- Currently only MacOS certificates
- (not a lot of expressed demand for Microsoft certificates)
- Ask your ISA contact to get started (Brian Maranville, Tim Snow...)

orsopy model language developments

- 1 minor release of orsopy
- Some overlap with data format group
- Artur Glavic/Max Skoda/Stephen Hall looking at developing model language for describing Lipids and Bilayers in ORSO file
- How this would interface with a downstream analysis package

sample:

model:

```
stack: "Si | DMPC | Bilayer{heads: PC, tails: DM} | biDMPC | (lbl3 | Bilayer{DMPC}) in H2O | Leaflet{rDMPC} | H2O"
```

sub_stacks:

DMPC:

sub_stack_class: Leaflet

heads: PC

tails:

composition:

DM: 0.9

PC: 0.1

heads_first: false

